BUSINESS ANALYTICS • EXECUTIVE PRESENTATION

EV Charging Infrastructure Analysis & Expansion Strategy

Business Analytics Dashboard

Tools Used
Microsoft Excel • Power Query • Power Pivot • DAX • Pivot Tables



Vinay Mehra

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SECTION 01 · FRAMING

Business Problem & Objectives

Problem Statement

With the rapid adoption of Electric Vehicles (EVs), Tata Power needs to evaluate whether its existing charging infrastructure can support future demand while ensuring profitability.
The objective is to analyze charging behavior, customer segments, infrastructure utilization, and financial performance to identify high-priority locations for future expansion.

Project Objectives

- Analyze charging demand and usage trends.
- Evaluate station utilization and infrastructure readiness.
- Understand customer charging behavior and revenue contribution.
- Assess project profitability using CAPEX and ROI.
- Recommend priority locations for future charging station deployment.

Dataset Overview

Dataset Summary

This project integrates multiple datasets to provide a comprehensive view of Tata Power's EV charging ecosystem.

Data Model

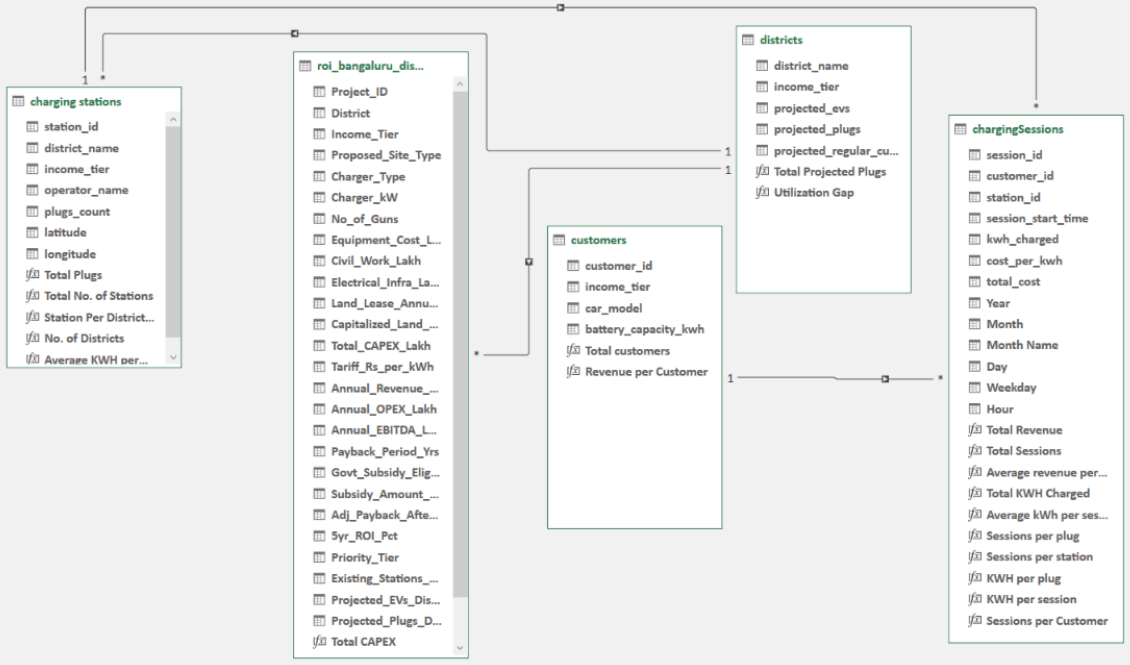
Relational Data Model using Power Pivot

Multiple One-to-Many Relationships

Fact & Dimension Tables



Data Model / ERD



Data Preparation & Methodology

Workflow / Process Overview



Data Cleaning

Before analysis, the data was cleaned, transformed, and modelled to ensure accurate business insights.

- Removed duplicate records
- Handled missing values
- Standardized data formats
- Extracted Year, Month, Day and Hour
- Built relationships using Power Pivot
- Created DAX Measures
- Developed Pivot Tables and
- Interactive Dashboards

Tools & Techniques

Power Query

Power Pivot

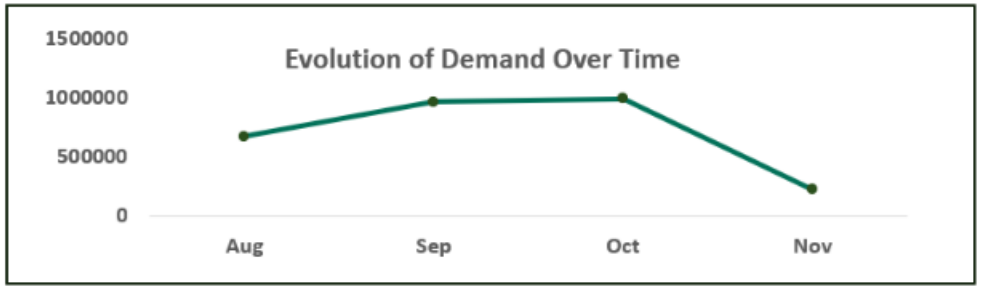
DAX

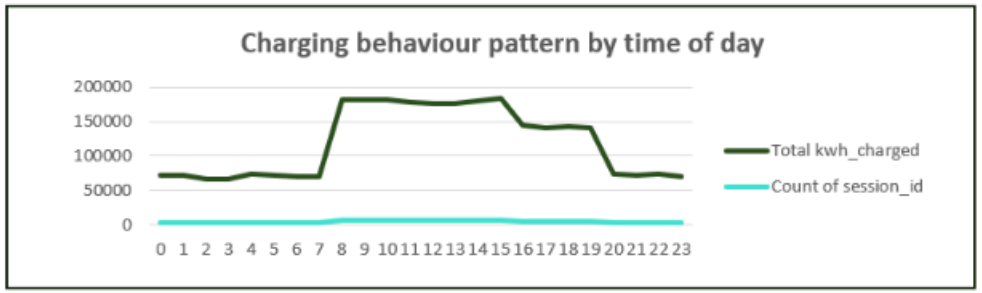
Dashboard 1 – Business & Usage Overview

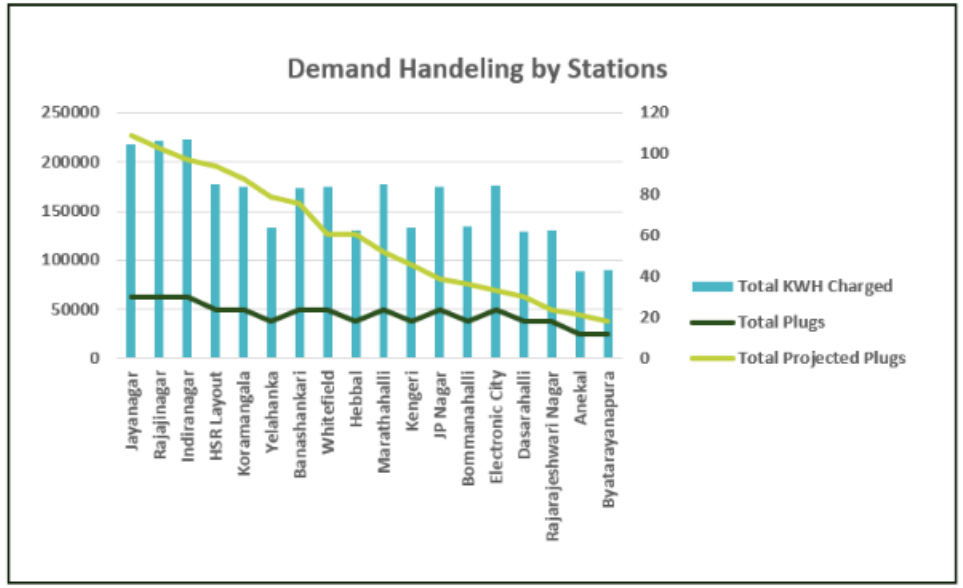
Business and Usage Overview

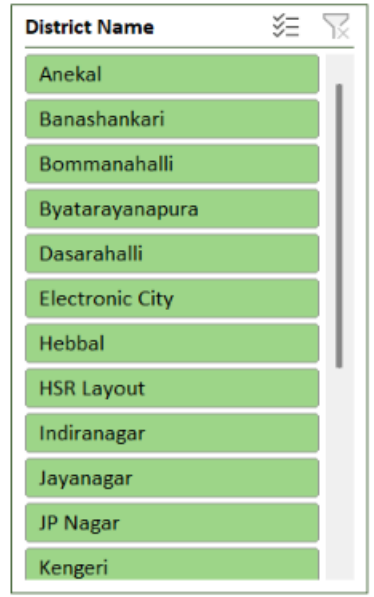
2 3 4

1,14,000.00	28,59,772	4,96,93,947	8,000	435.91	25.09
Total charging sessions	Total Energy Consumed	Total Revenue	Number of Active Customers	Average Revenue per session	Average KWH per session







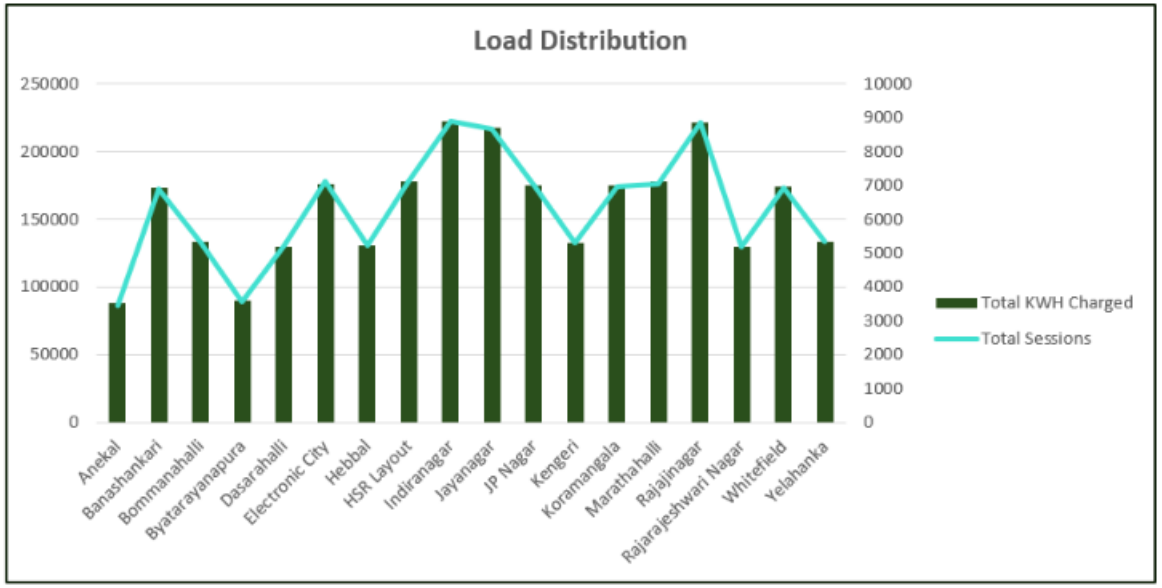


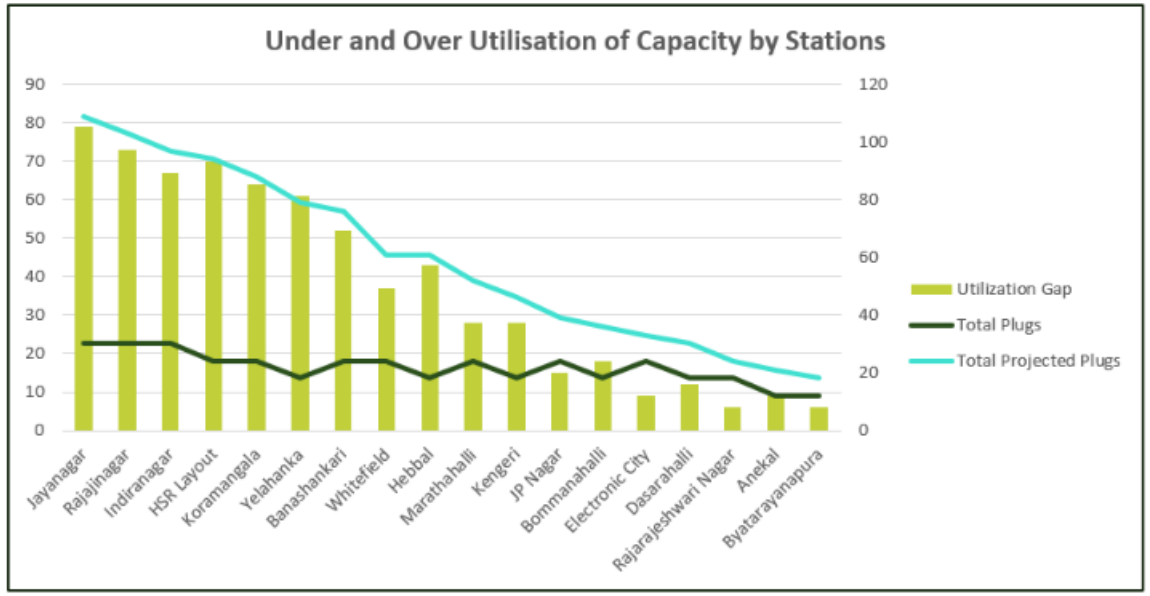
Dashboard 2 – Station Utilization & Infrastructure

Station Utilization and Operational Efficiency

1 3 4

1,754	292	25.09	3.61	292
Number of sessions per station	Average utilization per station	Average kWh per station	District wise station density	Sessions per plug ratio





Dashboard 3 – Customer & Demand Behaviour

Customer and Demand Behavior

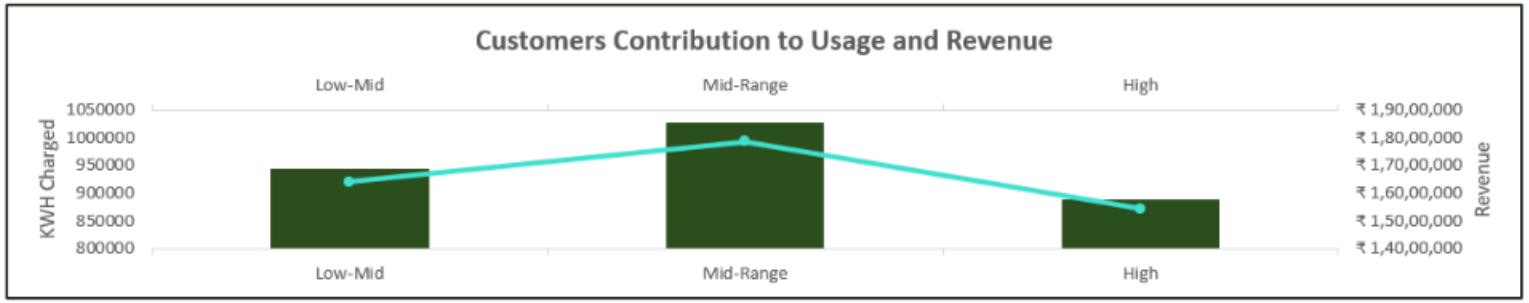
1 2 4

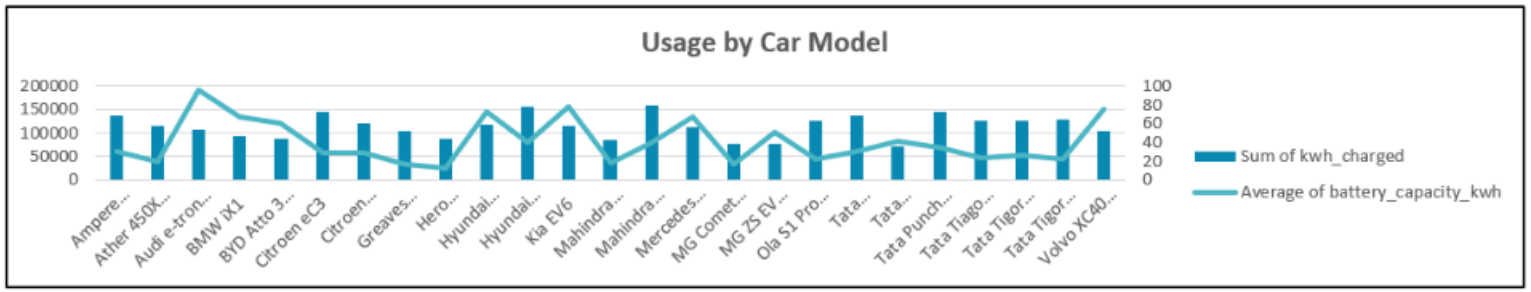
8,000
No. of Customers

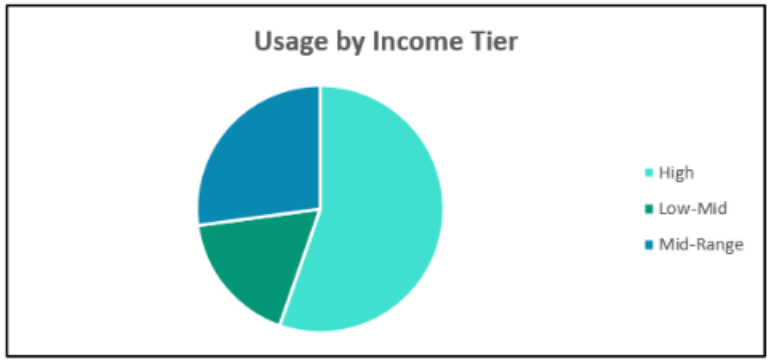
14.25
Sessions per Customer

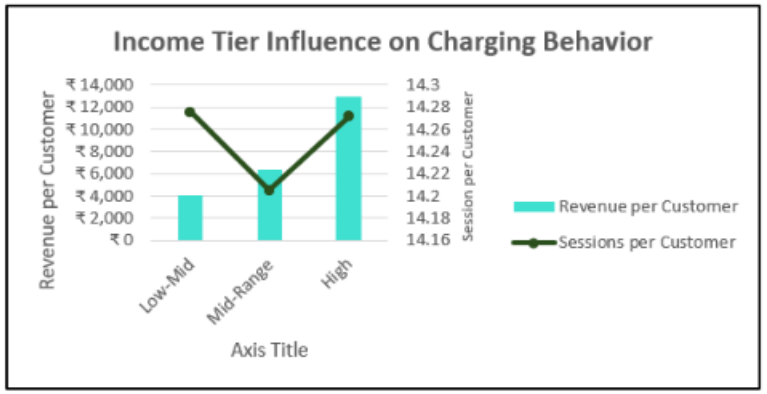
357.47
Average KWH per Customer

6,212
Revenue per Customer







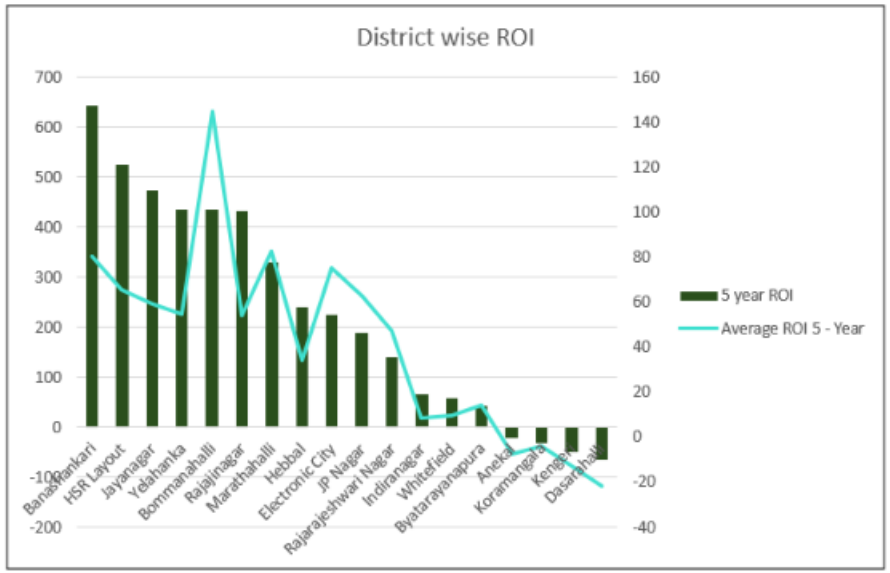


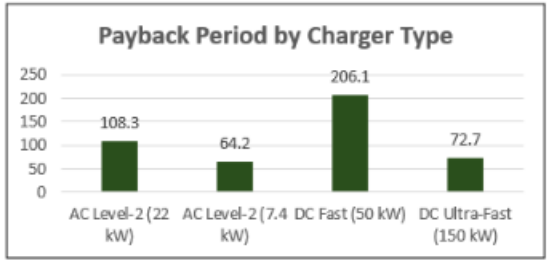
Dashboard 4 – Investment & Profitability

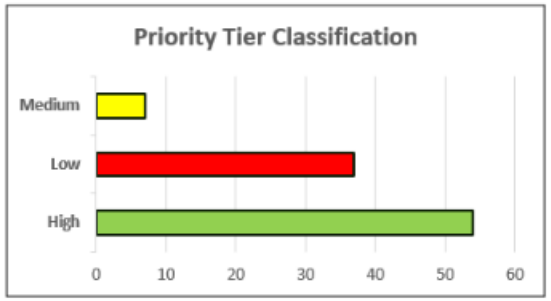
Financial Performance and Expansion Strategy (ROI Focus)

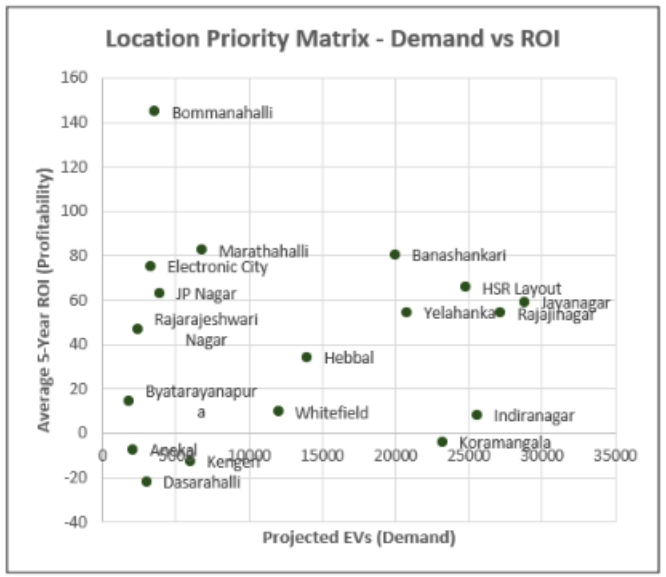
1 2 3

142	4,629	3,239	4.34	4.61	41.43
Total CAPEX Per Project	Annual Revenue	EBITDA	Payback Period (With Subsidy)	Payback Period before Subsidy	5 Year ROI %









Prioritised Locations

Immediate Expansion
- Jayanagar
- HSR Layout
- Banashankari

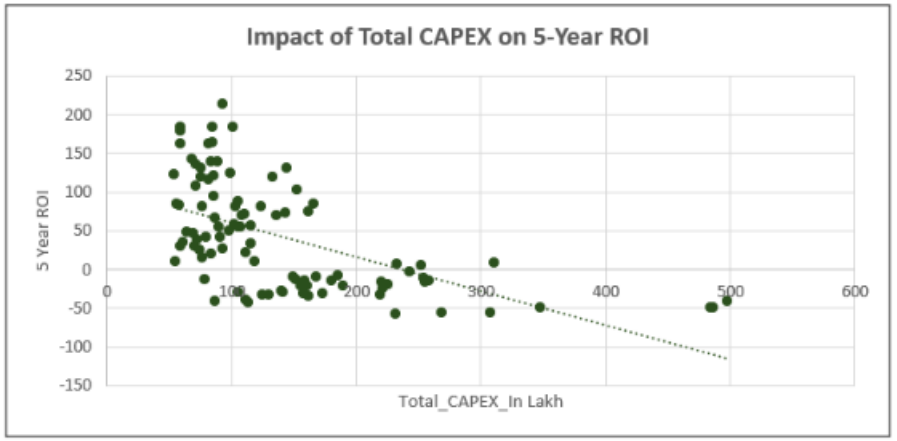
Investigate
- Koramangala
- Indiranagar

Future Expansion
- Bommanahalli
- Marathahalli

Low Priority
- Dasarahalli
- Anekal
- Kengeri

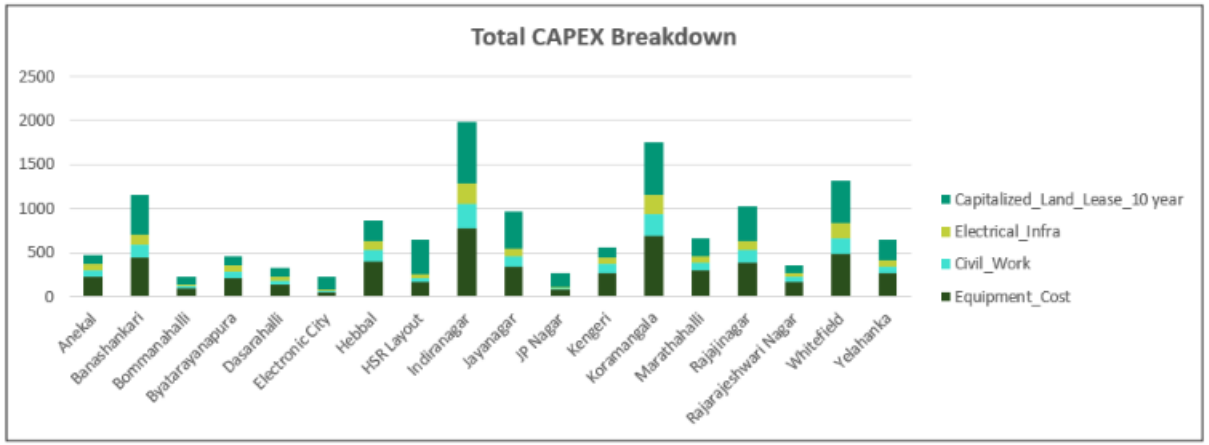


Dashboard 4.1 – Investment & Profitability



Projects with moderate capital investment generally achieve stronger ROI, whereas several high-CAPEX projects generate low or even negative returns. This indicates that larger investments alone do not guarantee profitability, and project performance depends on efficient utilization, operating costs, and revenue generation. **CAPEX between ₹60–150 Lakhs have ROI 100–200%. This suggests that several medium-investment projects generate significantly higher returns than very expensive projects.**

Review Required
CAPEX around ₹300 Lakhs ₹500 Lakhs have ROI around -50% -30% 0%. This suggests that Higher investment does not necessarily result in higher profitability.



Projected EV Demand
35200

Existing Stations
65

Current infrastructure consists of 65 charging stations, while projected EV demand is 35,200 vehicles. This indicates that future demand is expected to grow significantly and additional charging infrastructure will be required to support EV adoption.

CAPEX Insights

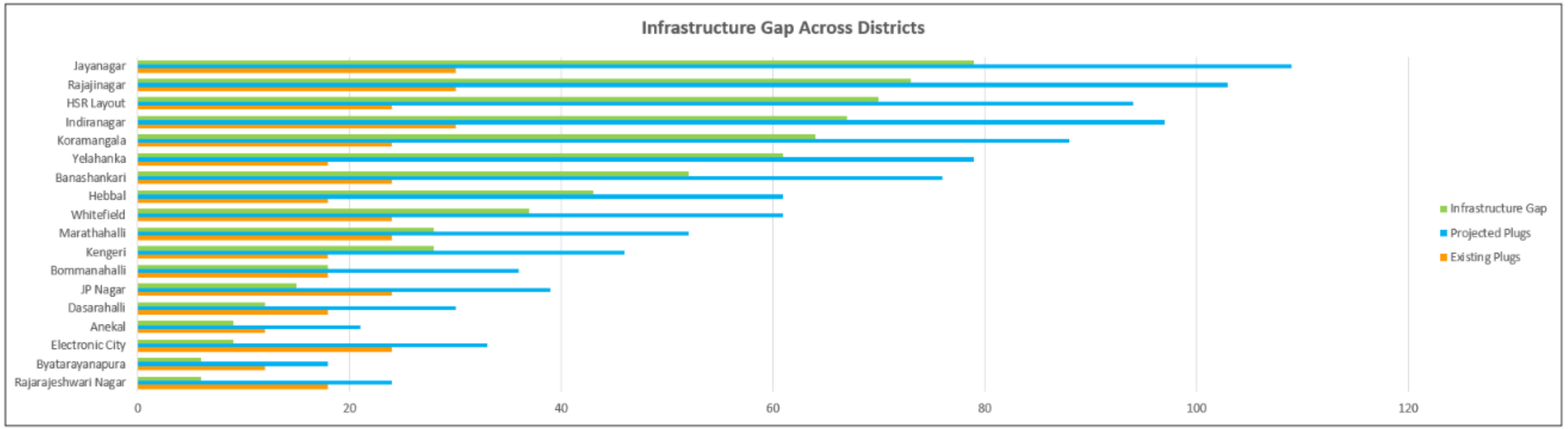
Equipment Cost
Largest contributor to Total CAPEX across most districts.

Infrastructure Costs
Civil Work and Electrical Infrastructure form the second-largest investment components.

Land Lease
Smaller share of CAPEX but contributes to long-term operating costs.

Investment Focus
Optimize equipment and infrastructure costs to improve project profitability.

Dashboard 4.2 – Investment & Profitability



Key Insights

- Current infrastructure provides **390 charging plugs** against a projected requirement of **1,067 plugs.**
- This leaves a shortfall of **677 plugs,** equivalent to approximately **113 additional charging stations** (assuming 6 plugs per station).
- **Jayanagar, Rajajinagar, HSR Layout, Indiranagar, and Koramangala** show the largest infrastructure gaps and should be prioritized for future expansion.

SECTION 08 · SYNTHESIS

Key Business Insights

01

EV charging demand is steadily increasing across districts.

02

Existing charging infrastructure is insufficient to meet projected demand.

03

Equipment Cost is the largest contributor to capital investment.

04

High-income customers generate the highest revenue.

05

Higher CAPEX generally reduces project profitability.

06

Jayanagar, HSR Layout and Banashankari are the highest-priority locations for expansion.

SECTION 09 · ROADMAP

Recommendations

Implementation Roadmap

[Phase 1]
Short-term

Expand High-Priority Locations

- *Expand charging infrastructure in Jayanagar, HSR Layout, and Banashankari.*
- *Add charging plugs in districts with the highest infrastructure gap.*
- *Improve utilization of underperforming charging stations through targeted marketing.*

[Phase 2]
Mid-term

Optimize Investment & Operations

- *Prioritize projects with high ROI and moderate CAPEX.*
- *Optimize equipment, civil work, and infrastructure costs to improve profitability.*
- *Continuously monitor station utilization and customer demand using operational dashboards.*

[Phase 3]
Long-term

Build a Data-Driven Expansion Strategy

- *Use predictive demand forecasting for future station deployment.*
- *Expand charging infrastructure in emerging high-demand districts.*
- *Continuously review customer behavior, profitability, and utilization to support strategic investment decisions.*

Future Strategy: *Adopt a data-driven expansion model by combining demand forecasting, infrastructure planning, customer analytics, and financial performance to maximize charging network utilization and long-term return on investment.*

THANK YOU

This project demonstrates how data analytics can support strategic decision-making in EV charging infrastructure planning. By combining operational, customer, and financial analysis, the dashboard identifies high-priority expansion opportunities while balancing infrastructure investment and profitability.

Vinay Mehra